

MASTER ASSIGNMENT REPORTS COLLECTION — TRAFFIC SIGNAL SYNCHRONIZATION

This document contains the individual project assignment reports for all 10 group members.

MATHEMATICS PROJECT ASSIGNMENT REPORT

PROJECT TITLE: Traffic Signal Synchronization System Using Chinese Remainder Theorem

COURSE: Number Theory & Discrete Mathematics Application

CLASS: S3 CSE B

STUDENT NAME: Joel Duke

ROLL NO: 51

ROLE / FUNCTION: Mathematical Research & Problem Formulation

1. ABSTRACT

The Chinese Remainder Theorem (CRT) is a fundamental result in number theory that determines a unique integer solution x for a system of linear congruences with pairwise coprime moduli. While often taught abstractly, CRT has direct applications in urban traffic signal synchronization, digital telecommunications, and parallel computer architectures.

This project demonstrates CRT using both a physical working traffic signal model and an interactive digital web application. By modeling traffic signals as modular timing cycles, the project illustrates how independent light timers (3 min, 5 min, 7 min) align to create a continuous 'Green Wave' for vehicles at Minute 23.

2. INTRODUCTION & OBJECTIVES

When traffic signals operate on independent timer cycles along a main avenue, drivers frequently encounter red lights. Traffic engineers solve this by calculating when all signals will simultaneously show green.

Project Objectives:

1. To understand the mathematical principles of the Chinese Remainder Theorem (CRT).
 2. To differentiate between Least Common Multiple (LCM) and CRT when remainder offsets are present.
 3. To model traffic signal timing cycles (3 min, 5 min, 7 min) as modular congruences.
 4. To build a physical working model and an interactive web simulator for signal synchronization.
 5. To provide clear visual and audio feedback when a candidate time satisfies all signal modulo conditions.
-

3. MATHEMATICAL FORMULATION (CRT vs. LCM)

3.1 The Traffic Signal Equations

Suppose three traffic signals have cycle lengths of 3, 5, and 7 minutes respectively:

- Signal 1: $x \equiv 2 \pmod{3}$ (green 2 minutes ago)
- Signal 2: $x \equiv 3 \pmod{5}$ (green 3 minutes ago)
- Signal 3: $x \equiv 2 \pmod{7}$ (green 2 minutes ago)

3.2 Why Simple LCM Cannot Solve This Problem

- LCM: Used only when all signals start at Minute 0 with zero remainder offset ($x \equiv 0$).
- CRT: Required when remainder offsets (2, 3, 2) are distinct and non-zero.

3.3 CRT Solution Procedure

1. Total Product: $M = 3 \times 5 \times 7 = 105$.

2. Partial Products: $M_1 = 35$, $M_2 = 21$, $M_3 = 15$.

3. Modular Inverses:

- $35 * y_1 \equiv 1 \pmod{3} \Rightarrow 2 * y_1 \equiv 1 \pmod{3} \Rightarrow y_1 = 2$
- $21 * y_2 \equiv 1 \pmod{5} \Rightarrow 1 * y_2 \equiv 1 \pmod{5} \Rightarrow y_2 = 1$
- $15 * y_3 \equiv 1 \pmod{7} \Rightarrow 1 * y_3 \equiv 1 \pmod{7} \Rightarrow y_3 = 1$

4. Compute x:

- $x = (2 \times 35 \times 2) + (3 \times 21 \times 1) + (2 \times 15 \times 1)$
- $x = 140 + 63 + 30 = 233$
- $x \equiv 233 \pmod{105} \equiv 23 \pmod{105}$

The unique green-wave solution within $[0, 104]$ is Minute 23.

4. DIGITAL IMPLEMENTATION & USER CONTROLS

The web application models the traffic signal system digitally using React, TypeScript, and Framer Motion.





Interactive User Controls:

1. Candidate Timer Wheel: Adjusts the candidate time value x (e.g. set to 23).
2. Intersection Signal Monitors (3 min, 5 min, 7 min): Shows real-time remainder values ($x \bmod m_i$) and green light indicator status.
3. 'Evaluate Synchronization' Button: Tests whether all 3 intersections display green lights simultaneously.
4. Green-Wave Status Panel: Displays full green-wave confirmation when all 3 congruences are satisfied ($x = 23$).

5. SOURCE CODE IMPLEMENTATION (ALGORITHM PAGE)

The following complete TypeScript module (src/utlis/crt.ts) implements the Extended Euclidean Algorithm and Modular Inverses to solve the CRT system:

```
export function getRemainder(value: number, modulus: number): number {
    return ((value % modulus) + modulus) % modulus;
}

function extendedGcd(a: number, b: number): [number, number, number] {
    if (b === 0) return [a, 1, 0];
    const [g, x1, y1] = extendedGcd(b, a % b);
    return [g, y1, x1 - Math.floor(a / b) * y1];
}

function modInverse(a: number, m: number): number {
    const [, x] = extendedGcd(getRemainder(a, m), m);
    return getRemainder(x, m);
}

/** Solves  $x \equiv \text{remainders}[i] \pmod{\text{moduli}[i]}$  for pairwise-coprime moduli */
export function solveCRT(moduli: number[], remainders: number[]): number {
    const product = moduli.reduce((acc, m) => acc * m, 1);
    let result = 0;
    for (let i = 0; i < moduli.length; i++) {
        const mi = moduli[i];
        const ri = getRemainder(remainders[i], mi);
        const partial = product / mi;
```

```
        const inverse = modInverse(partial, mi);
        result += ri * partial * inverse;
    }
    return getRemainder(result, product);
}

export function checkRemainders(x: number, moduli: number[], remainders: number[]): boolean[] {
    return moduli.map((m, i) => getRemainder(x, m) === getRemainder(remainders[i], m));
}
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6. CONCLUSION

The Traffic Signal Synchronization System provides an intuitive, practical bridge between abstract number theory and urban infrastructure engineering. By applying the Chinese Remainder Theorem to traffic signals operating on independent cycles (3 min, 5 min, 7 min), the project demonstrates how non-zero remainder offsets determine the exact minute (Minute 23) required to establish a continuous 'Green Wave' for vehicles.

Both the physical working model and the interactive web application successfully visualize modular congruences, offering an accessible learning tool that highlights the real-world power of modular arithmetic in signal timing, telecommunications, and digital computing.

7. INDIVIDUAL CONTRIBUTION

Name: Joel Duke

Class: S3 CSE B

Roll No: 51

Role: Mathematical Research & Problem Formulation

Contribution Summary:

I researched the mathematical foundation of linear congruences. I formulated the traffic signal green-wave equations for 3, 5, and 7-minute cycles and calculated the unique solution bounds modulo 105.

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MATHEMATICS PROJECT ASSIGNMENT REPORT

PROJECT TITLE: Traffic Signal Synchronization System Using Chinese Remainder Theorem

COURSE: Number Theory & Discrete Mathematics Application

CLASS: S3 CSE B

STUDENT NAME: Joel Geo Manuel

ROLL NO: 52

ROLE / FUNCTION: Presentation & Showcase Lead

1. ABSTRACT

The Chinese Remainder Theorem (CRT) is a fundamental result in number theory that determines a unique integer solution x for a system of linear congruences with pairwise coprime moduli. While often taught abstractly, CRT has direct applications in urban traffic signal synchronization, digital telecommunications, and parallel computer architectures.

This project demonstrates CRT using both a physical working traffic signal model and an interactive digital web application. By modeling traffic signals as modular timing cycles, the project illustrates how independent light timers (3 min, 5 min, 7 min) align to create a continuous 'Green Wave' for vehicles at Minute 23.

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4. Compute x:
 - $x = (2 \times 35 \times 2) + (3 \times 21 \times 1) + (2 \times 15 \times 1)$
 - $x = 140 + 63 + 30 = 233$
 - $x \equiv 233 \pmod{105} \equiv 23 \pmod{105}$

The unique green-wave solution within $[0, 104]$ is Minute 23.

4. DIGITAL IMPLEMENTATION & USER CONTROLS

The web application models the traffic signal system digitally using React, TypeScript, and Framer Motion.



Interactive User Controls:

1. Candidate Timer Wheel: Adjusts the candidate time value x (e.g. set to 23).
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    if (b === 0) return [a, 1, 0];
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    return [g, y1, x1 - Math.floor(a / b) * y1];
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function modInverse(a: number, m: number): number {
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export function solveCRT(moduli: number[], remainders: number[]): number {
    const product = moduli.reduce((acc, m) => acc * m, 1);
    let result = 0;
    for (let i = 0; i < moduli.length; i++) {
        const mi = moduli[i];
        const ri = getRemainder(remainders[i], mi);
        const partial = product / mi;
        const inverse = modInverse(partial, mi);
        result += ri * partial * inverse;
    }
    return getRemainder(result, product);
}

export function checkRemainders(x: number, moduli: number[], remainders: number[]): boolean[] {
    return moduli.map((m, i) => getRemainder(x, m) === getRemainder(remainders[i], m));
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The Traffic Signal Synchronization System provides an intuitive, practical bridge between abstract number theory and urban infrastructure engineering. By applying the Chinese Remainder Theorem to traffic signals operating on independent cycles (3 min, 5 min, 7 min), the project demonstrates how non-zero remainder offsets determine the exact minute (Minute 23) required to establish a continuous 'Green Wave' for vehicles.

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7. INDIVIDUAL CONTRIBUTION

Name: Joel Geo Manuel

Class: S3 CSE B

Roll No: 52

Role: Presentation & Showcase Lead

Contribution Summary:

I designed the presentation structure, demo walkthrough flow, and visual slides. I formulated the showcase script explaining the Chinese Remainder Theorem through the traffic signal analogy for live audience demonstrations.

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MATHEMATICS PROJECT ASSIGNMENT REPORT

PROJECT TITLE: Traffic Signal Synchronization System Using Chinese Remainder Theorem

COURSE: Number Theory & Discrete Mathematics Application

CLASS: S3 CSE B

STUDENT NAME: Johaan Sam

ROLL NO: 53

ROLE / FUNCTION: Frontend & State Engine Developer

1. ABSTRACT

The Chinese Remainder Theorem (CRT) is a fundamental result in number theory that determines a unique integer solution x for a system of linear congruences with pairwise coprime moduli. While often taught abstractly, CRT has direct applications in urban traffic signal synchronization, digital telecommunications, and parallel computer architectures.

This project demonstrates CRT using both a physical working traffic signal model and an interactive digital web application. By modeling traffic signals as modular timing cycles, the project illustrates how independent light timers (3 min, 5 min, 7 min) align to create a continuous 'Green Wave' for vehicles at Minute 23.

2. INTRODUCTION & OBJECTIVES

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Project Objectives:

1. To understand the mathematical principles of the Chinese Remainder Theorem (CRT).
 2. To differentiate between Least Common Multiple (LCM) and CRT when remainder offsets are present.
 3. To model traffic signal timing cycles (3 min, 5 min, 7 min) as modular congruences.
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-

3. MATHEMATICAL FORMULATION (CRT vs. LCM)

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1. Total Product: $M = 3 \times 5 \times 7 = 105$.
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4. Compute x:

- $x = (2 \times 35 \times 2) + (3 \times 21 \times 1) + (2 \times 15 \times 1)$
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- $x \equiv 233 \pmod{105} \equiv 23 \pmod{105}$

The unique green-wave solution within $[0, 104]$ is Minute 23.

4. DIGITAL IMPLEMENTATION & USER CONTROLS

The web application models the traffic signal system digitally using React, TypeScript, and Framer Motion.





Interactive User Controls:

1. Candidate Timer Wheel: Adjusts the candidate time value x (e.g. set to 23).
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5. SOURCE CODE IMPLEMENTATION (ALGORITHM PAGE)

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export function solveCRT(moduli: number[], remainders: number[]): number {
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    let result = 0;
    for (let i = 0; i < moduli.length; i++) {
        const mi = moduli[i];
        const ri = getRemainder(remainders[i], mi);
        const partial = product / mi;
```

```
    const inverse = modInverse(partial, mi);
    result += ri * partial * inverse;
  }
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  return moduli.map((m, i) => getRemainder(x, m) === getRemainder(remainders[i], m));
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6. CONCLUSION

The Traffic Signal Synchronization System provides an intuitive, practical bridge between abstract number theory and urban infrastructure engineering. By applying the Chinese Remainder Theorem to traffic signals operating on independent cycles (3 min, 5 min, 7 min), the project demonstrates how non-zero remainder offsets determine the exact minute (Minute 23) required to establish a continuous 'Green Wave' for vehicles.

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7. INDIVIDUAL CONTRIBUTION

Name: Johaan Sam

Class: S3 CSE B

Roll No: 53

Role: Frontend & State Engine Developer

Contribution Summary:

I developed the frontend state logic and component structure. Working with Johan Geo, I built the interactive controls for the timer wheel and state transition hooks managing traffic light synchronization and green-wave status indicators.

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MATHEMATICS PROJECT ASSIGNMENT REPORT

PROJECT TITLE: Traffic Signal Synchronization System Using Chinese Remainder Theorem

COURSE: Number Theory & Discrete Mathematics Application

CLASS: S3 CSE B

STUDENT NAME: Johan Geo

ROLL NO: 54

ROLE / FUNCTION: Web Design & UI Architect

1. ABSTRACT

The Chinese Remainder Theorem (CRT) is a fundamental result in number theory that determines a unique integer solution x for a system of linear congruences with pairwise coprime moduli. While often taught abstractly, CRT has direct applications in urban traffic signal synchronization, digital telecommunications, and parallel computer architectures.

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3. MATHEMATICAL FORMULATION (CRT vs. LCM)

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4. Compute x:
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The unique green-wave solution within $[0, 104]$ is Minute 23.

4. DIGITAL IMPLEMENTATION & USER CONTROLS

The web application models the traffic signal system digitally using React, TypeScript, and Framer Motion.



Interactive User Controls:

1. Candidate Timer Wheel: Adjusts the candidate time value x (e.g. set to 23).
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7. INDIVIDUAL CONTRIBUTION

Name: Johan Geo

Class: S3 CSE B

Roll No: 54

Role: Web Design & UI Architect

Contribution Summary:

I designed the website interface, visual layout, and traffic signal dashboard theme. I created the interactive modular dial components and responsive styling using Tailwind CSS, ensuring clear visual feedback for remainder checkmarks and signal light states.

MATHEMATICS PROJECT ASSIGNMENT REPORT

PROJECT TITLE: Traffic Signal Synchronization System Using Chinese Remainder Theorem

COURSE: Number Theory & Discrete Mathematics Application

CLASS: S3 CSE B

STUDENT NAME: Jose Alex

ROLL NO: 55

ROLE / FUNCTION: Mathematical Proofs & Solution Verification

1. ABSTRACT

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/** Solves  $x \equiv \text{remainders}[i] \pmod{\text{moduli}[i]}$  for pairwise-coprime moduli */
export function solveCRT(moduli: number[], remainders: number[]): number {
    const product = moduli.reduce((acc, m) => acc * m, 1);
    let result = 0;
    for (let i = 0; i < moduli.length; i++) {
        const mi = moduli[i];
        const ri = getRemainder(remainders[i], mi);
        const partial = product / mi;
```

```
        const inverse = modInverse(partial, mi);
        result += ri * partial * inverse;
    }
    return getRemainder(result, product);
}

export function checkRemainders(x: number, moduli: number[], remainders: number[]): boolean[] {
    return moduli.map((m, i) => getRemainder(x, m) === getRemainder(remainders[i], m));
}
```

6. CONCLUSION

The Traffic Signal Synchronization System provides an intuitive, practical bridge between abstract number theory and urban infrastructure engineering. By applying the Chinese Remainder Theorem to traffic signals operating on independent cycles (3 min, 5 min, 7 min), the project demonstrates how non-zero remainder offsets determine the exact minute (Minute 23) required to establish a continuous 'Green Wave' for vehicles.

Both the physical working model and the interactive web application successfully visualize modular congruences, offering an accessible learning tool that highlights the real-world power of modular arithmetic in signal timing, telecommunications, and digital computing.

7. INDIVIDUAL CONTRIBUTION

Name: Jose Alex

Class: S3 CSE B

Roll No: 55

Role: Mathematical Proofs & Solution Verification

Contribution Summary:

I investigated Chinese Remainder Theorem proofs, verifying solution uniqueness within the range [0, 104]. I tested edge-case remainder combinations to ensure mathematical accuracy across all puzzle configurations.

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MATHEMATICS PROJECT ASSIGNMENT REPORT

PROJECT TITLE: Traffic Signal Synchronization System Using Chinese Remainder Theorem

COURSE: Number Theory & Discrete Mathematics Application

CLASS: S3 CSE B

STUDENT NAME: Joseph Alex

ROLL NO: 56

ROLE / FUNCTION: Hardware Sourcing & Build Testing

1. ABSTRACT

The Chinese Remainder Theorem (CRT) is a fundamental result in number theory that determines a unique integer solution x for a system of linear congruences with pairwise coprime moduli. While often taught abstractly, CRT has direct applications in urban traffic signal synchronization, digital telecommunications, and parallel computer architectures.

This project demonstrates CRT using both a physical working traffic signal model and an interactive digital web application. By modeling traffic signals as modular timing cycles, the project illustrates how independent light timers (3 min, 5 min, 7 min) align to create a continuous 'Green Wave' for vehicles at Minute 23.

2. INTRODUCTION & OBJECTIVES

When traffic signals operate on independent timer cycles along a main avenue, drivers frequently encounter red lights. Traffic engineers solve this by calculating when all signals will simultaneously show green.

Project Objectives:

1. To understand the mathematical principles of the Chinese Remainder Theorem (CRT).
 2. To differentiate between Least Common Multiple (LCM) and CRT when remainder offsets are present.
 3. To model traffic signal timing cycles (3 min, 5 min, 7 min) as modular congruences.
 4. To build a physical working model and an interactive web simulator for signal synchronization.
 5. To provide clear visual and audio feedback when a candidate time satisfies all signal modulo conditions.
-

3. MATHEMATICAL FORMULATION (CRT vs. LCM)

3.1 The Traffic Signal Equations

Suppose three traffic signals have cycle lengths of 3, 5, and 7 minutes respectively:

- Signal 1: $x \equiv 2 \pmod{3}$ (green 2 minutes ago)
- Signal 2: $x \equiv 3 \pmod{5}$ (green 3 minutes ago)
- Signal 3: $x \equiv 2 \pmod{7}$ (green 2 minutes ago)

3.2 Why Simple LCM Cannot Solve This Problem

- LCM: Used only when all signals start at Minute 0 with zero remainder offset ($x \equiv 0$).
- CRT: Required when remainder offsets (2, 3, 2) are distinct and non-zero.

3.3 CRT Solution Procedure

1. Total Product: $M = 3 \times 5 \times 7 = 105$.
2. Partial Products: $M_1 = 35$, $M_2 = 21$, $M_3 = 15$.
3. Modular Inverses:
 - $35 * y_1 \equiv 1 \pmod{3} \Rightarrow 2 * y_1 \equiv 1 \pmod{3} \Rightarrow y_1 = 2$
 - $21 * y_2 \equiv 1 \pmod{5} \Rightarrow 1 * y_2 \equiv 1 \pmod{5} \Rightarrow y_2 = 1$
 - $15 * y_3 \equiv 1 \pmod{7} \Rightarrow 1 * y_3 \equiv 1 \pmod{7} \Rightarrow y_3 = 1$
4. Compute x:
 - $x = (2 \times 35 \times 2) + (3 \times 21 \times 1) + (2 \times 15 \times 1)$
 - $x = 140 + 63 + 30 = 233$
 - $x \equiv 233 \pmod{105} \equiv 23 \pmod{105}$

The unique green-wave solution within $[0, 104]$ is Minute 23.

4. DIGITAL IMPLEMENTATION & USER CONTROLS

The web application models the traffic signal system digitally using React, TypeScript, and Framer Motion.



Interactive User Controls:

1. Candidate Timer Wheel: Adjusts the candidate time value x (e.g. set to 23).
2. Intersection Signal Monitors (3 min, 5 min, 7 min): Shows real-time remainder values ($x \bmod m_i$) and green light indicator status.
3. 'Evaluate Synchronization' Button: Tests whether all 3 intersections display green lights simultaneously.
4. Green-Wave Status Panel: Displays full green-wave confirmation when all 3 congruences are satisfied ($x = 23$).

5. SOURCE CODE IMPLEMENTATION (ALGORITHM PAGE)

The following complete TypeScript module (src/utis/crt.ts) implements the Extended Euclidean Algorithm and Modular Inverses to solve the CRT system:

```
export function getRemainder(value: number, modulus: number): number {
    return ((value % modulus) + modulus) % modulus;
}

function extendedGcd(a: number, b: number): [number, number, number] {
    if (b === 0) return [a, 1, 0];
    const [g, x1, y1] = extendedGcd(b, a % b);
    return [g, y1, x1 - Math.floor(a / b) * y1];
}

function modInverse(a: number, m: number): number {
    const [, x] = extendedGcd(getRemainder(a, m), m);
    return getRemainder(x, m);
}

/** Solves  $x \equiv \text{remainders}[i] \pmod{\text{moduli}[i]}$  for pairwise-coprime moduli */
export function solveCRT(moduli: number[], remainders: number[]): number {
    const product = moduli.reduce((acc, m) => acc * m, 1);
    let result = 0;
    for (let i = 0; i < moduli.length; i++) {
        const mi = moduli[i];
        const ri = getRemainder(remainders[i], mi);
        const partial = product / mi;
        const inverse = modInverse(partial, mi);
        result += ri * partial * inverse;
    }
    return getRemainder(result, product);
}

export function checkRemainders(x: number, moduli: number[], remainders: number[]): boolean[] {
    return moduli.map((m, i) => getRemainder(x, m) === getRemainder(remainders[i], m));
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6. CONCLUSION

The Traffic Signal Synchronization System provides an intuitive, practical bridge between abstract number theory and urban infrastructure engineering. By applying the Chinese Remainder Theorem to traffic signals operating on independent cycles (3 min, 5 min, 7 min), the project demonstrates how non-zero remainder offsets determine the exact minute (Minute 23) required to establish a continuous 'Green Wave' for vehicles.

Both the physical working model and the interactive web application successfully visualize modular congruences, offering an accessible learning tool that highlights the real-world power of modular arithmetic in signal timing, telecommunications, and digital computing.

7. INDIVIDUAL CONTRIBUTION

Name: Joseph Alex

Class: S3 CSE B

Roll No: 56

Role: Hardware Sourcing & Build Testing

Contribution Summary:

I determined the hardware part requirements for the model, sourced necessary components, and executed physical build testing. I also contributed to application use-cases, exploring how modular arithmetic aligns traffic signal cycles in real-world urban infrastructure.

MATHEMATICS PROJECT ASSIGNMENT REPORT

PROJECT TITLE: Traffic Signal Synchronization System Using Chinese Remainder Theorem

COURSE: Number Theory & Discrete Mathematics Application

CLASS: S3 CSE B

STUDENT NAME: Joseph J

ROLL NO: 57

ROLE / FUNCTION: Component Procurement & Application Ideas

1. ABSTRACT

The Chinese Remainder Theorem (CRT) is a fundamental result in number theory that determines a unique integer solution x for a system of linear congruences with pairwise coprime moduli. While often taught abstractly, CRT has direct applications in urban traffic signal synchronization, digital telecommunications, and parallel computer architectures.

This project demonstrates CRT using both a physical working traffic signal model and an interactive digital web application. By modeling traffic signals as modular timing cycles, the project illustrates how independent light timers (3 min, 5 min, 7 min) align to create a continuous 'Green Wave' for vehicles at Minute 23.

2. INTRODUCTION & OBJECTIVES

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Project Objectives:

1. To understand the mathematical principles of the Chinese Remainder Theorem (CRT).
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4. Compute x :

$$x = (2 \times 35 \times 2) + (3 \times 21 \times 1) + (2 \times 15 \times 1)$$

$$x = 140 + 63 + 30 = 233$$

$$x \equiv 233 \pmod{105} \equiv 23 \pmod{105}$$

The unique green-wave solution within $[0, 104]$ is Minute 23.

4. DIGITAL IMPLEMENTATION & USER CONTROLS

The web application models the traffic signal system digitally using React, TypeScript, and Framer Motion.





Interactive User Controls:

1. Candidate Timer Wheel: Adjusts the candidate time value x (e.g. set to 23).
2. Intersection Signal Monitors (3 min, 5 min, 7 min): Shows real-time remainder values ($x \bmod m_i$) and green light indicator status.
3. 'Evaluate Synchronization' Button: Tests whether all 3 intersections display green lights simultaneously.
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    if (b === 0) return [a, 1, 0];
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function modInverse(a: number, m: number): number {
    const [, x] = extendedGcd(getRemainder(a, m), m);
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/** Solves  $x \equiv \text{remainders}[i] \pmod{\text{moduli}[i]}$  for pairwise-coprime moduli */
export function solveCRT(moduli: number[], remainders: number[]): number {
    const product = moduli.reduce((acc, m) => acc * m, 1);
    let result = 0;
    for (let i = 0; i < moduli.length; i++) {
        const mi = moduli[i];
        const ri = getRemainder(remainders[i], mi);
        const partial = product / mi;
```

```
        const inverse = modInverse(partial, mi);
        result += ri * partial * inverse;
    }
    return getRemainder(result, product);
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export function checkRemainders(x: number, moduli: number[], remainders: number[]): boolean[] {
    return moduli.map((m, i) => getRemainder(x, m) === getRemainder(remainders[i], m));
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6. CONCLUSION

The Traffic Signal Synchronization System provides an intuitive, practical bridge between abstract number theory and urban infrastructure engineering. By applying the Chinese Remainder Theorem to traffic signals operating on independent cycles (3 min, 5 min, 7 min), the project demonstrates how non-zero remainder offsets determine the exact minute (Minute 23) required to establish a continuous 'Green Wave' for vehicles.

Both the physical working model and the interactive web application successfully visualize modular congruences, offering an accessible learning tool that highlights the real-world power of modular arithmetic in signal timing, telecommunications, and digital computing.

7. INDIVIDUAL CONTRIBUTION

Name: Joseph J

Class: S3 CSE B

Roll No: 57

Role: Component Procurement & Application Ideas

Contribution Summary:

I handled component selection, hardware procurement, and assembly testing alongside Joseph Alex. Additionally, I assisted in designing application scenarios and creating vector schematics to visually represent the internal signal timing mechanism for project documentation.

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MATHEMATICS PROJECT ASSIGNMENT REPORT

PROJECT TITLE: Traffic Signal Synchronization System Using Chinese Remainder Theorem

COURSE: Number Theory & Discrete Mathematics Application

CLASS: S3 CSE B

STUDENT NAME: Justin Joe Mathew

ROLL NO: 58

ROLE / FUNCTION: Team Lead & Working Model Engineer

1. ABSTRACT

The Chinese Remainder Theorem (CRT) is a fundamental result in number theory that determines a unique integer solution x for a system of linear congruences with pairwise coprime moduli. While often taught abstractly, CRT has direct applications in urban traffic signal synchronization, digital telecommunications, and parallel computer architectures.

This project demonstrates CRT using both a physical working traffic signal model and an interactive digital web application. By modeling traffic signals as modular timing cycles, the project illustrates how independent light timers (3 min, 5 min, 7 min) align to create a continuous 'Green Wave' for vehicles at Minute 23.

2. INTRODUCTION & OBJECTIVES

When traffic signals operate on independent timer cycles along a main avenue, drivers frequently encounter red lights. Traffic engineers solve this by calculating when all signals will simultaneously show green.

Project Objectives:

1. To understand the mathematical principles of the Chinese Remainder Theorem (CRT).
 2. To differentiate between Least Common Multiple (LCM) and CRT when remainder offsets are present.
 3. To model traffic signal timing cycles (3 min, 5 min, 7 min) as modular congruences.
 4. To build a physical working model and an interactive web simulator for signal synchronization.
 5. To provide clear visual and audio feedback when a candidate time satisfies all signal modulo conditions.
-

3. MATHEMATICAL FORMULATION (CRT vs. LCM)

3.1 The Traffic Signal Equations

Suppose three traffic signals have cycle lengths of 3, 5, and 7 minutes respectively:

- Signal 1: $x \equiv 2 \pmod{3}$ (green 2 minutes ago)
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1. Total Product: $M = 3 \times 5 \times 7 = 105$.
2. Partial Products: $M_1 = 35$, $M_2 = 21$, $M_3 = 15$.
3. Modular Inverses:
 - $35 * y_1 \equiv 1 \pmod{3} \Rightarrow 2 * y_1 \equiv 1 \pmod{3} \Rightarrow y_1 = 2$
 - $21 * y_2 \equiv 1 \pmod{5} \Rightarrow 1 * y_2 \equiv 1 \pmod{5} \Rightarrow y_2 = 1$
 - $15 * y_3 \equiv 1 \pmod{7} \Rightarrow 1 * y_3 \equiv 1 \pmod{7} \Rightarrow y_3 = 1$
4. Compute x:
 - $x = (2 \times 35 \times 2) + (3 \times 21 \times 1) + (2 \times 15 \times 1)$
 - $x = 140 + 63 + 30 = 233$
 - $x \equiv 233 \pmod{105} \equiv 23 \pmod{105}$

The unique green-wave solution within $[0, 104]$ is Minute 23.

4. DIGITAL IMPLEMENTATION & USER CONTROLS

The web application models the traffic signal system digitally using React, TypeScript, and Framer Motion.



Interactive User Controls:

1. Candidate Timer Wheel: Adjusts the candidate time value x (e.g. set to 23).
2. Intersection Signal Monitors (3 min, 5 min, 7 min): Shows real-time remainder values ($x \bmod m_i$) and green light indicator status.
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5. SOURCE CODE IMPLEMENTATION (ALGORITHM PAGE)

The following complete TypeScript module (src/utls/crt.ts) implements the Extended Euclidean Algorithm and Modular Inverses to solve the CRT system:

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export function getRemainder(value: number, modulus: number): number {
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    if (b === 0) return [a, 1, 0];
    const [g, x1, y1] = extendedGcd(b, a % b);
    return [g, y1, x1 - Math.floor(a / b) * y1];
}

function modInverse(a: number, m: number): number {
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    const product = moduli.reduce((acc, m) => acc * m, 1);
    let result = 0;
    for (let i = 0; i < moduli.length; i++) {
        const mi = moduli[i];
        const ri = getRemainder(remainders[i], mi);
        const partial = product / mi;
        const inverse = modInverse(partial, mi);
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6. CONCLUSION

The Traffic Signal Synchronization System provides an intuitive, practical bridge between abstract number theory and urban infrastructure engineering. By applying the Chinese Remainder Theorem to traffic signals operating on independent cycles (3 min, 5 min, 7 min), the project demonstrates how non-zero remainder offsets determine the exact minute (Minute 23) required to establish a continuous 'Green Wave' for vehicles.

Both the physical working model and the interactive web application successfully visualize modular congruences, offering an accessible learning tool that highlights the real-world power of modular arithmetic in signal timing, telecommunications, and digital computing.

7. INDIVIDUAL CONTRIBUTION

Name: Justin Joe Mathew

Class: S3 CSE B

Roll No: 58

Role: Team Lead & Working Model Engineer

Contribution Summary:

I served as Team Lead, guiding overall project execution and system architecture. I led the construction and logic development of the working traffic signal model, implementing the mathematical solver engine using the Extended Euclidean Algorithm to calculate modular multiplicative inverses for traffic light synchronization.

MATHEMATICS PROJECT ASSIGNMENT REPORT

PROJECT TITLE: Traffic Signal Synchronization System Using Chinese Remainder Theorem

COURSE: Number Theory & Discrete Mathematics Application

CLASS: S3 CSE B

STUDENT NAME: Jyothika Prakash

ROLL NO: 59

ROLE / FUNCTION: Pedagogical Systems & Educational Design

1. ABSTRACT

The Chinese Remainder Theorem (CRT) is a fundamental result in number theory that determines a unique integer solution x for a system of linear congruences with pairwise coprime moduli. While often taught abstractly, CRT has direct applications in urban traffic signal synchronization, digital telecommunications, and parallel computer architectures.

This project demonstrates CRT using both a physical working traffic signal model and an interactive digital web application. By modeling traffic signals as modular timing cycles, the project illustrates how independent light timers (3 min, 5 min, 7 min) align to create a continuous 'Green Wave' for vehicles at Minute 23.

2. INTRODUCTION & OBJECTIVES

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Project Objectives:

1. To understand the mathematical principles of the Chinese Remainder Theorem (CRT).
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3. MATHEMATICAL FORMULATION (CRT vs. LCM)

3.1 The Traffic Signal Equations

Suppose three traffic signals have cycle lengths of 3, 5, and 7 minutes respectively:

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$$x = 140 + 63 + 30 = 233$$

$$x \equiv 233 \pmod{105} \equiv 23 \pmod{105}$$

The unique green-wave solution within $[0, 104]$ is Minute 23.

4. DIGITAL IMPLEMENTATION & USER CONTROLS

The web application models the traffic signal system digitally using React, TypeScript, and Framer Motion.





Interactive User Controls:

1. Candidate Timer Wheel: Adjusts the candidate time value x (e.g. set to 23).
2. Intersection Signal Monitors (3 min, 5 min, 7 min): Shows real-time remainder values ($x \bmod m_i$) and green light indicator status.
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7. INDIVIDUAL CONTRIBUTION

Name: Jyothika Prakash

Class: S3 CSE B

Roll No: 59

Role: Pedagogical Systems & Educational Design

Contribution Summary:

I explored real-world CRT applications and designed the interactive hint system and step-by-step mathematical explanations to help users understand remainder arithmetic without complex jargon.

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MATHEMATICS PROJECT ASSIGNMENT REPORT

PROJECT TITLE: Traffic Signal Synchronization System Using Chinese Remainder Theorem

COURSE: Number Theory & Discrete Mathematics Application

CLASS: S3 CSE B

STUDENT NAME: Jyothis Liju

ROLL NO: 60

ROLE / FUNCTION: Working Model Co-Developer

1. ABSTRACT

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4. Compute x:
 - $x = (2 \times 35 \times 2) + (3 \times 21 \times 1) + (2 \times 15 \times 1)$
 - $x = 140 + 63 + 30 = 233$
 - $x \equiv 233 \pmod{105} \equiv 23 \pmod{105}$

The unique green-wave solution within $[0, 104]$ is Minute 23.

4. DIGITAL IMPLEMENTATION & USER CONTROLS

The web application models the traffic signal system digitally using React, TypeScript, and Framer Motion.



Interactive User Controls:

1. Candidate Timer Wheel: Adjusts the candidate time value x (e.g. set to 23).
2. Intersection Signal Monitors (3 min, 5 min, 7 min): Shows real-time remainder values ($x \bmod m_i$) and green light indicator status.
3. 'Evaluate Synchronization' Button: Tests whether all 3 intersections display green lights simultaneously.
4. Green-Wave Status Panel: Displays full green-wave confirmation when all 3 congruences are satisfied ($x = 23$).

5. SOURCE CODE IMPLEMENTATION (ALGORITHM PAGE)

The following complete TypeScript module (src/utlis/crt.ts) implements the Extended Euclidean Algorithm and Modular Inverses to solve the CRT system:

```
export function getRemainder(value: number, modulus: number): number {
    return ((value % modulus) + modulus) % modulus;
}

function extendedGcd(a: number, b: number): [number, number, number] {
    if (b === 0) return [a, 1, 0];
    const [g, x1, y1] = extendedGcd(b, a % b);
    return [g, y1, x1 - Math.floor(a / b) * y1];
}

function modInverse(a: number, m: number): number {
    const [, x] = extendedGcd(getRemainder(a, m), m);
    return getRemainder(x, m);
}

/** Solves  $x \equiv \text{remainders}[i] \pmod{\text{moduli}[i]}$  for pairwise-coprime moduli */
export function solveCRT(moduli: number[], remainders: number[]): number {
    const product = moduli.reduce((acc, m) => acc * m, 1);
    let result = 0;
    for (let i = 0; i < moduli.length; i++) {
        const mi = moduli[i];
        const ri = getRemainder(remainders[i], mi);
        const partial = product / mi;
        const inverse = modInverse(partial, mi);
        result += ri * partial * inverse;
    }
    return getRemainder(result, product);
}

export function checkRemainders(x: number, moduli: number[], remainders: number[]): boolean[] {
    return moduli.map((m, i) => getRemainder(x, m) === getRemainder(remainders[i], m));
}
```

6. CONCLUSION

The Traffic Signal Synchronization System provides an intuitive, practical bridge between abstract number theory and urban infrastructure engineering. By applying the Chinese Remainder Theorem to traffic signals operating on independent cycles (3 min, 5 min, 7 min), the project demonstrates how non-zero remainder offsets determine the exact minute (Minute 23) required to establish a continuous 'Green Wave' for vehicles.

Both the physical working model and the interactive web application successfully visualize modular congruences, offering an accessible learning tool that highlights the real-world power of modular arithmetic in signal timing, telecommunications, and digital computing.

7. INDIVIDUAL CONTRIBUTION

Name: Jyothis Liju

Class: S3 CSE B

Roll No: 60

Role: Working Model Co-Developer

Contribution Summary:

I collaborated directly with Justin Joe Mathew on constructing the working traffic signal model. I assisted in component assembly, signal alignment testing, and verifying electrical/mechanical clearance to ensure reliable execution when all modulo remainder conditions were met.

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